

Stability of various systems — Solution

X18 (2018) — English translation

A1

1.50 pts

Express the angular acceleration of rod $\ddot{\varphi}$ in terms of φ , l and g . Show that the time dependence of the inclination angle φ is given by $\varphi = Ae^{t/\tau} + Be^{-t/\tau}$, where A and B are constants depending on the initial position and initial angular velocity of the rod, and τ is the characteristic time of fall. Express τ in terms of l and g .

The moment equation for the rod about the hinge is:

$$I = M\ddot{\varphi},$$

where $I = \frac{1}{3}ml^2$ is the moment of inertia of the rod about its end, and $M = mg \cdot (\frac{1}{2}l \sin \varphi) \simeq \frac{1}{2}mgl\varphi$ is the moment of gravity.

Let's check that $\varphi = Ae^{t/\tau} + Be^{-t/\tau}$ is a solution to the resulting equation: $\ddot{\varphi} = \frac{1}{\tau^2} (Ae^{t/\tau} + Be^{-t/\tau}) = \frac{\varphi(t)}{\tau^2}$.

Answer: \textbf{Answer: } $\ddot{\varphi} = \frac{3g}{2l}\varphi$.

A2

0.50 pts

Let the boy now try to support a long thin stick in an upright position on his hand. As soon as the stick begins to fall, for example, to the left, he moves his hand to the left an even greater distance so that the center of mass of the stick is to the right of its fulcrum. As a result, the moment of gravity begins to rotate the stick to the right, reducing its speed of rotation directed to the left. Estimate how long a stick the boy can hold vertically if his reaction time is approximately $\tau_r = 0,2$ s. (Reaction time is the time delay between the command sent by the brain to the hands and the corresponding movement of the hands.)

The boy's reaction time τ_r must be of the order of the characteristic fall time τ . Using this, we find the length of the stick: $l_r = \tau_r^2 \frac{3g}{2}$.

Answer: \textbf{Answer: }

$$l_r = \tau_r^2 \frac{3g}{2} = 0.59 \text{ m}$$

A3

0.50 pts

Humans and birds maintain a standing position in a similar way. They move the fulcrum (the point at the bottom of the foot where normal ground reaction force is applied), for example, changing the angle between the foot and leg to counteract the fall of the upper body. A small bird of height $l = 6$ cm can stand on its feet. Estimate her reaction time from above.

The bird's reaction time τ_b must be of the order of the characteristic fall time τ . Using this, we find the bird's reaction time: $\tau_b = \sqrt{\frac{2l_b}{3g}}$.

Answer: \textbf{Answer: }

$$\tau_b = \sqrt{\frac{2l_b}{3g}} = 0.065 \text{ s}$$